

Wachirawit Piyaprapapan

AI Engineer | Model Serving, Neural Architectures, Forecasting

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Education

Chulalongkorn University, B.Eng. in Electrical Engineering – Bangkok, Thailand

Aug 2022 – May 2026

- GPAX: 3.44 (Second-class honours)
- Coursework: Data Science, Data Engineering, Estimation, Statistical Learning, Optimization
- Capstone: Generative Video-Based Sky Image Forecasting for Thai Sky Images (SkyGPT, VQ-VAE)

Skills

Languages: Python, SQL, Bash

AI / ML: PyTorch, FastAPI, OpenCV, YOLO, spaCy

Systems: Docker, PostgreSQL, SQLAlchemy, Grafana

Data: Spark, Airflow, Pandas, Scikit-learn

Experience

AI Engineer Part Time, Hobbit Technologies – Bangkok, Thailand

Feb 2026 – present

- Engineered a production-ready RAG and generation pipeline with LiteLLM and FastAPI, replacing free-form text with Pydantic-validated structured outputs for reliable downstream parsing.
- Integrated spaCy and Oxford 5000-derived data into a modular Next.js 15, FastAPI, and PostgreSQL platform to power CEFR-aligned content generation and learner feedback.

AI Engineer Intern, Hobbit Technologies – Bangkok, Thailand

June 2025 – Aug 2025

- Built an internal computer-vision annotation platform, reducing labeling cost by around 20k Baht for faster YOLO iteration.
- Implemented logging and monitoring pipelines with daily reports, improving observability and operational readiness.

Projects

Generative Video-Based Sky Image Forecasting for Thai Sky Images

- Optimized the VQ-VAE stage with perceptual loss, EMA codebook updates, and architectural tuning to improve reconstruction fidelity for sky forecasting.
- Surpassed baseline models at peak validation PSNR of 37.32 dB (+16%) and produced stable discrete latents for downstream temporal prediction.

AI-Powered English Learning Platform

- Built structured quiz and CEFR-reading generation flows with FastAPI, PostgreSQL, SQLAlchemy, LiteLLM, and spaCy.
- Used typed outputs instead of free-form text to improve inference reliability and deployment readiness.

On-Demand Delivery Data Platform & Decision Intelligence System

- Deployed delay prediction services with FastAPI and scheduled refreshes to keep inference aligned with time-series drift.
- Packaged the pipeline with Docker and production checks to reduce friction between experimentation and serving.